

CS & IT ENGINEERING

Programming in C



Strings
Lec- 02



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TOPICS TO BE
COVERED



Srrings-II



String $\Rightarrow$ Address

char arr[10] = "Pankaj";

Using 1 char array $\Rightarrow$ 1 string

Using 2 char array $\Rightarrow$ 2 string

char arr[10] = "Pankaj";
char arr1[10] = "Neeraj";
char arr2[10] = "Poonam";



2-D array of
characters

char name[3][10] =

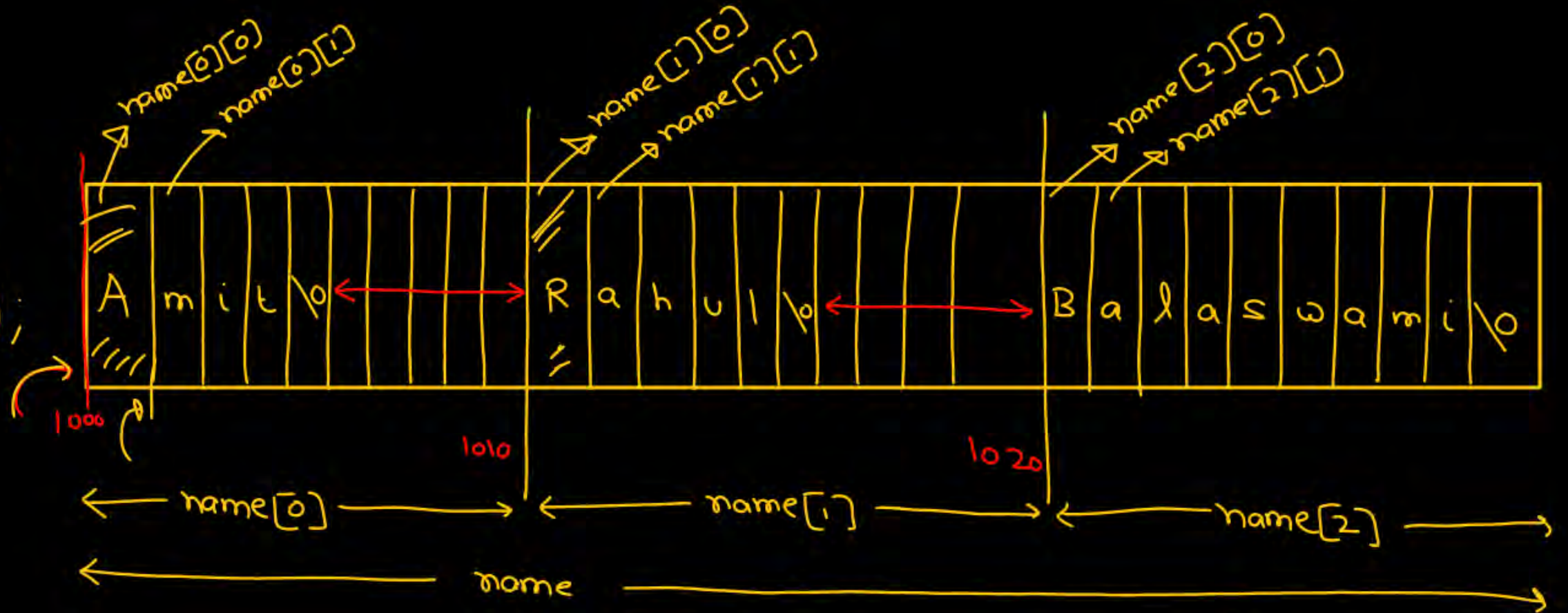
{ "Pankaj", "Neeraj", "Poonam" };

char name[3][10] = {"Amit", "Rahul", "Balaswami"};

name[0] $\Rightarrow$ ✓ Add/ele

name[0] $\Rightarrow$ &name[0][0]

Amit { printf("/s", name[0]);
printf(name[0]);



name[1] $\Rightarrow$ Address

name[1] = &name[1][0]

Rahul { printf("/s", name[1]);
printf("/s", &name[1][0]);
printf(name[1]);
printf(&name[1][0]);

①

*name + 1 $\Rightarrow$ * &name[0] + 1
 $\Rightarrow$ name[0] + 1
 $\Rightarrow$ &name[0][0] + 1
 $\Rightarrow$ &name[0][1]

char name[3][10] = {"Amit", "Rahul", "Balaswami"};

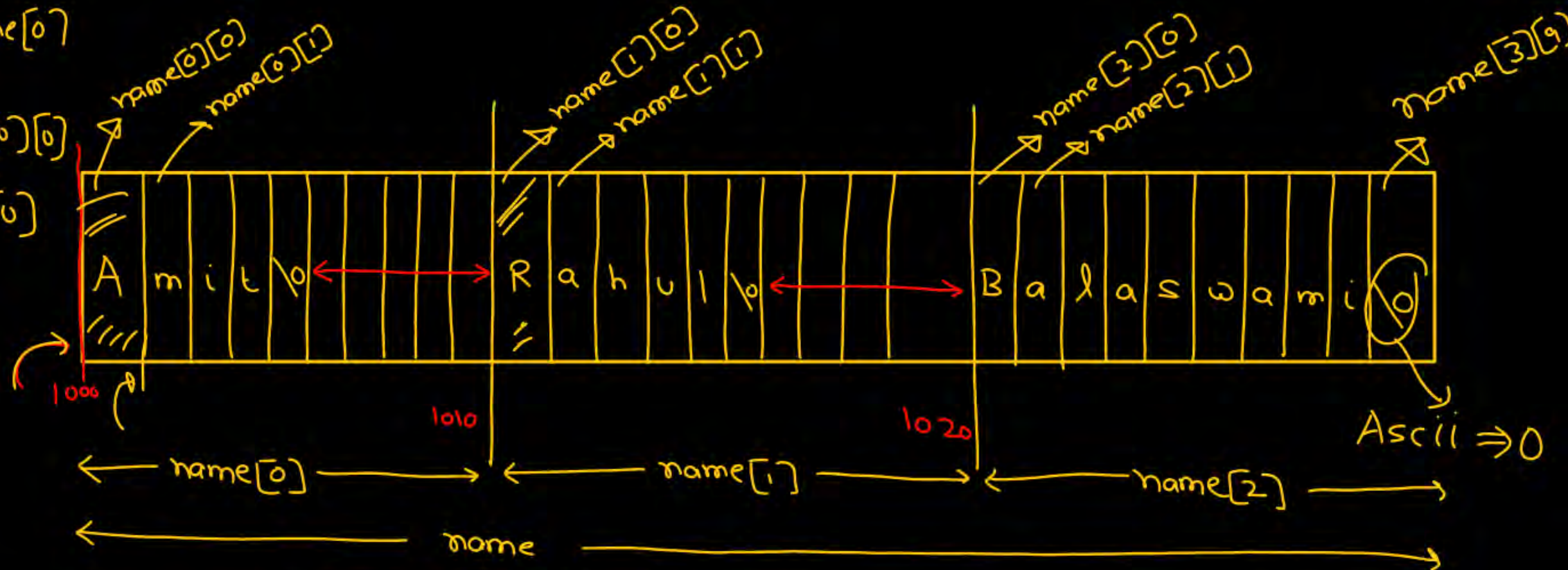
4* name

→ 4* name[0]

⇒ 4* name[0][0]

⇒ name[0][0]

printf("/c", name);



name[0] = "Pankaj"; {value required}

char * name = "Pankaj" ;

char * name1 = "Neeraj" ;

char * name2 = "Poonam" ;



using array of pointer to char

char * name[3]

using 3 (pointer to char)

⇒ 3 string

$\text{name} \Rightarrow \&\text{name}[0] \Rightarrow \text{Mem. loc. } 52$

$\ast \text{name} \Rightarrow \ast \cancel{\&\text{name}[0]} \Rightarrow \ast (\text{Mem. loc. } 52)$

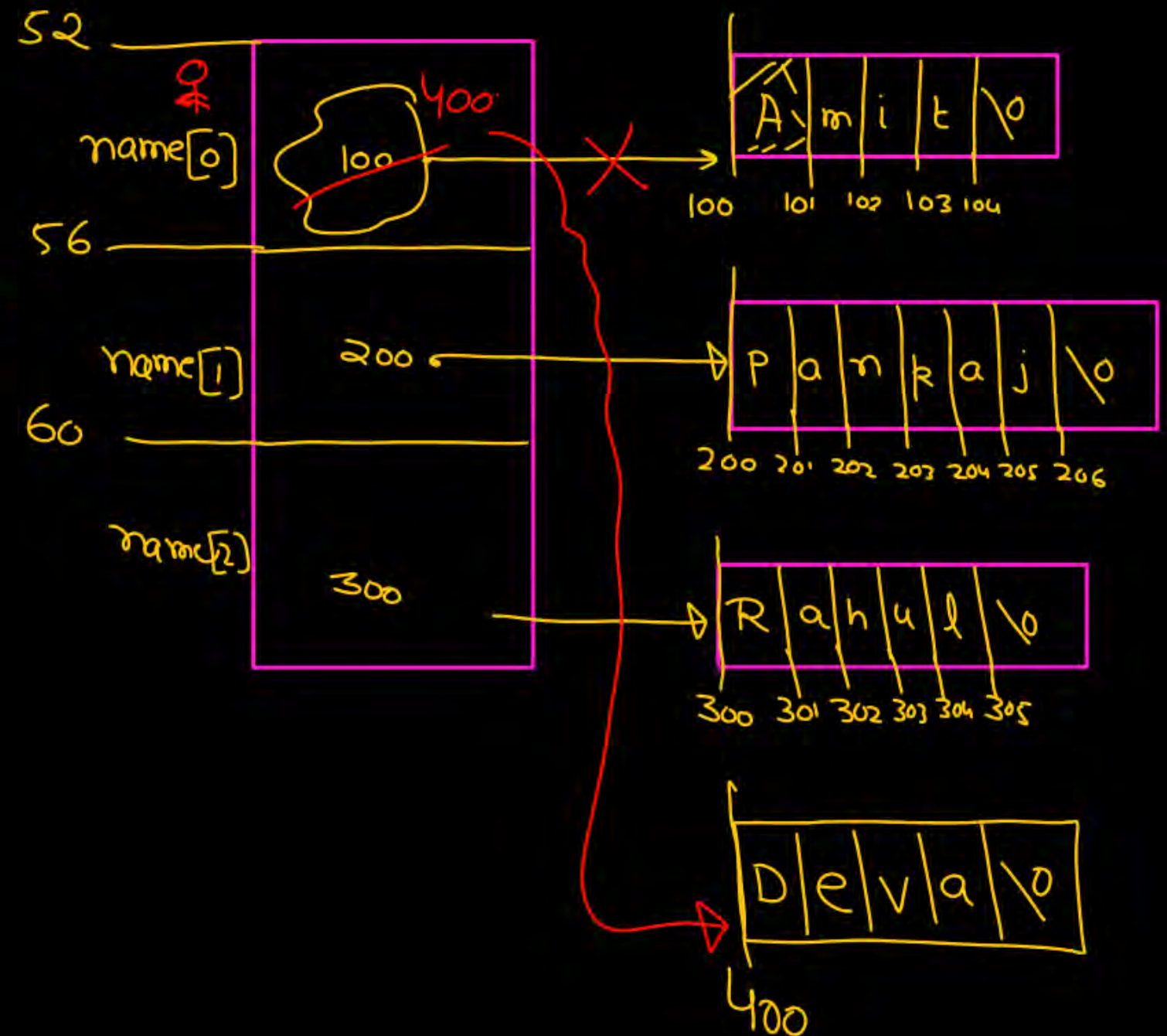
$\ast \text{name} \Rightarrow \text{name}[0] \Rightarrow \text{Mem. loc. } 100$

$\text{printf}("/s", \ast \text{name});$
 $\text{printf}("/s", \text{name}[0]);$

$\text{printf}("/s", \ast \text{name} + 1);$ mit

$\text{name}[0] = \text{"Deva"};$ ✓

$\text{char } \ast \text{name}[3] = \{ \text{"Amit"}, \text{"Panraj"}, \text{"Rahul"} \};$



In-built function

string.h

1) strlen : Returns the length of the string.
(No. of char/symbols in string but \0 is not counted)

unsigned int

```
#include <string.h>
```

```
void main() {
```

```
    char *p = "Neeraj";
```

```
    char arr[] = "Pankaj";
```

```
    printf("/u", sizeof(arr));
```

```
    printf("/u", strlen(p));
```

```
    printf("/u", strlen(arr));
```

```
}
```

```
void main() {
```

```
    char *p = "amit\0 Pankaj\0";
```

```
    printf("/u", strlen(p));
```

```
}
```

Can we implement our own strlen

can not be modified by this function

```
void main(){
```

```
    char *p = "Neeraj";
```

```
    char arr[] = "Pankaj";
```

```
    printf("/u", my_strlen(p));
```

```
    printf("/u", my_strlen(arr));
```

```
}
```

```
unsigned int my_strlen(const char *str)
```

```
{
```

```
    unsigned int c = 0;
```

```
    while (*str != '\0')
```

```
    {
```

```
        c++;
```

```
        str++;
```

```
    }
```

```
    return c;
```

```
}
```


2. strcpy

To copy a string

char name[20];

Invalid name = "Pankaj";

char *
return
copied string

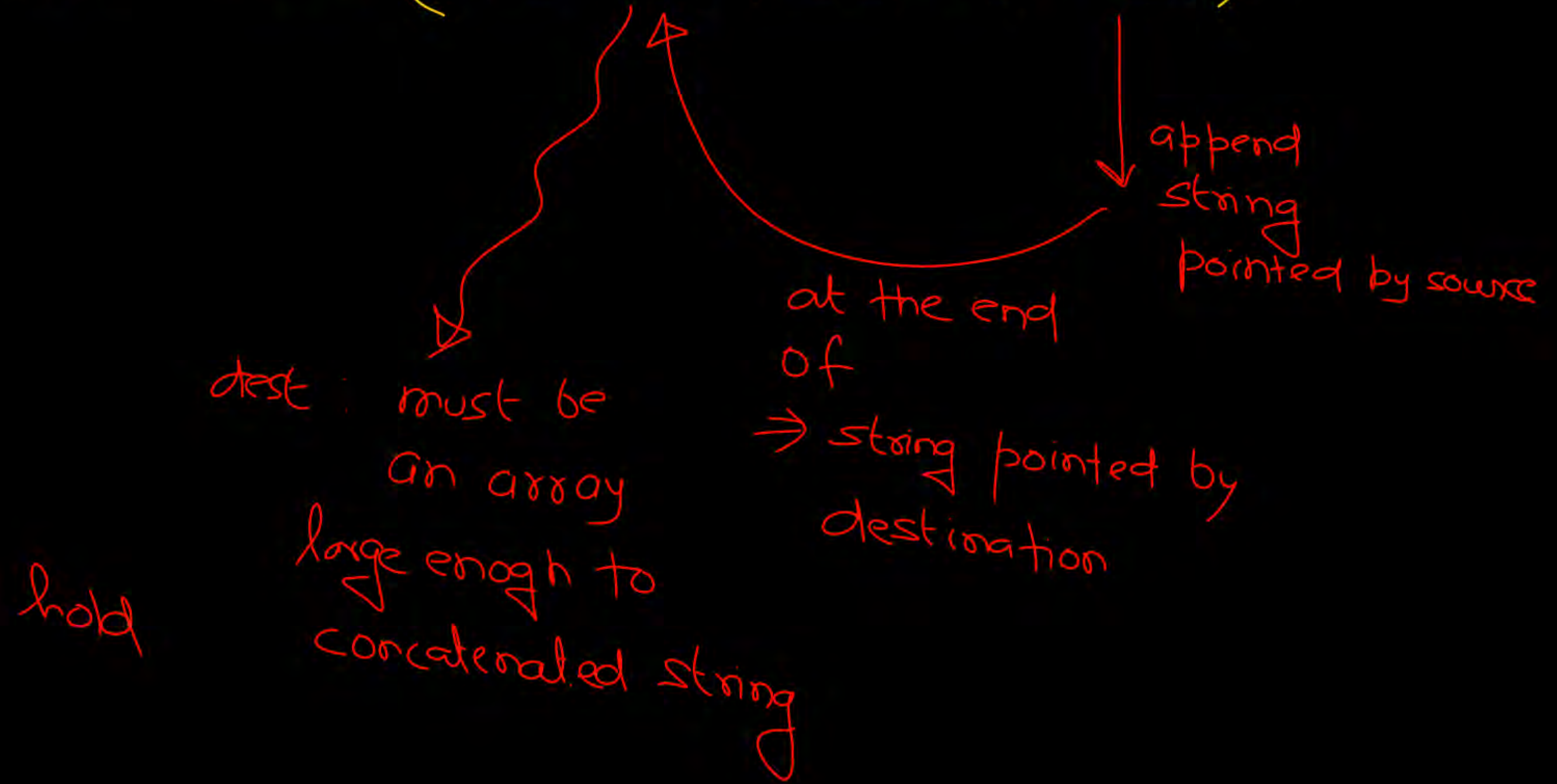
strcpy(char *destination,

const char *source);

It copies the string pointed by source pointer to the string memory pointed by destination pointer.

[size of destination buffer must be large enough to store source string.]
Also copy \0

char * strcat (char *dest , const char *source)




```
[ char arr[20];  
  strcat(arr, "Pankaj"); ]
```

```
[ char arr[20] = "Pankaj";  
  strcat(arr, "sharma");  
  printf("%s", arr); ]
```

Pankajsharma

strcmp

used to compare 2 strings

(i) 0

`strcmp("Ram", "Ram");` $\rightarrow 0$

`char *p = "Ram";`

`char arr[] = "Ram";`

`strcmp(p, arr);` $\rightarrow 0$

(ii)

+ve

`strcmp("Rani", "Ramu");`
'n' > 'm'

(iii)

-ve

`strcmp("Ramu", "Rani")`

~~Ramu~~, ~~Rani~~
↑ ↑

First unmatched
char. is found

'm' < 'n'
Ascii
value Ascii
 value

[strlen
strcpy
strcat
strcmp]

Try to
implement

.....

$\exists$ PM
└─> [structure]

```
char arr[10];
```

```
strcpy(arr, "Pankaj");
```

```
printf("%s", arr);
```

P	a	n	k	a	j	\0			
---	---	---	---	---	---	----	--	--	--

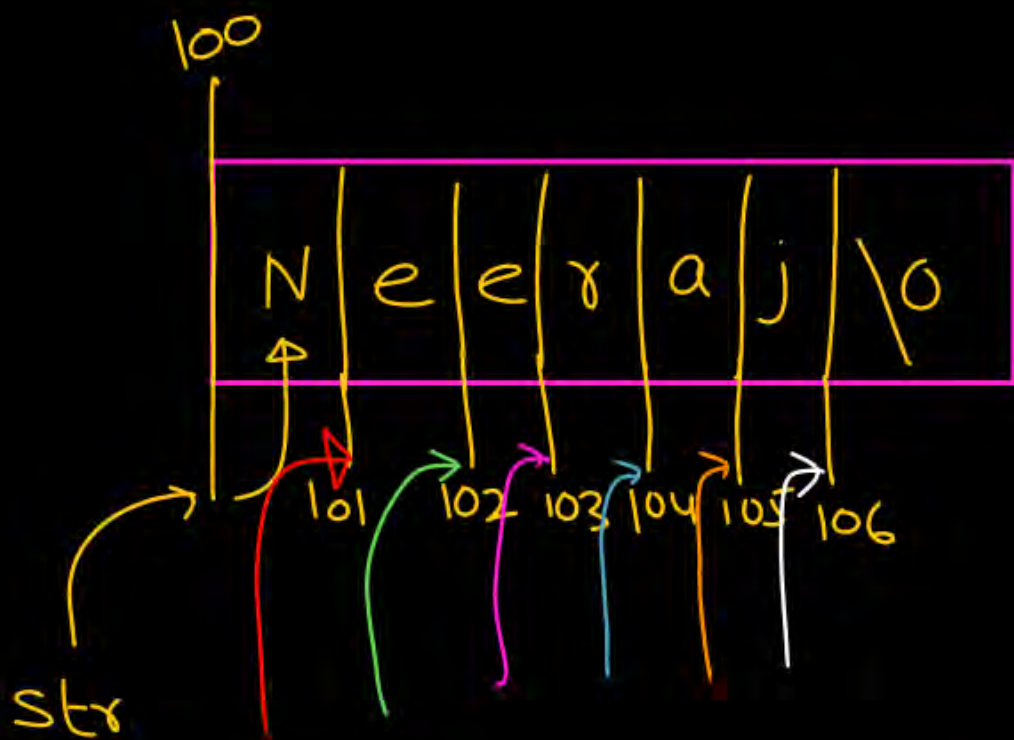
char *p;

strcpy(p, "Neeraj");

X Error


```
char * my_strcpy ( char * destination, const char * source )
{
    char * ptr = destination;
    if ( destination == NULL )
        return NULL;

    while ( *source != '\0' )
    {
        *ptr = *source;
        ptr++;
        source++;
    }
    *ptr = '\0'; ⇒ Explicitly
    return destination;
}
```



~~5~~ ~~4~~ ~~3~~
~~6~~ ~~2~~
~~+~~
 $C = \emptyset;$

*str $\Rightarrow$ is it NULL $\Rightarrow$ NO
C++;

Move to next add

str++;

vii) *str $\Rightarrow$ is it NULL $\Rightarrow$ Yes
STOP

(iv) *str $\Rightarrow$ is it NULL $\Rightarrow$ NO 'r'
C++
str++

(v) *str $\Rightarrow$ is it NULL $\Rightarrow$ NO 'a'
C++
str++

(vi) *str $\Rightarrow$ is it NULL $\Rightarrow$ NO 'j'
C++
str++

*str $\Rightarrow$ is it NULL $\Rightarrow$ NO 'e'
C++;

Move to next add.
str++

*str $\Rightarrow$ is it NULL $\Rightarrow$ NO 'e'
C++
Move to next add
str++;

